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Institution: University of Florida
Course: LIN 6923: Special Topics – All Computational Models Are Wrong
Assignment: Project Proposal
Topic: Subjunctive Moods

The subjunctive mood (SM) in linguistics is characterized as a non-factual action that has yet to occur (Negi and Buitelaar, 2015; Amirqulova, 2026). It has some form of suggestion, wish, possibility, desire, advice, recommendation, or warning (Negi and Buitelaar, 2015; Anand et al., 2019; Duy, 2019).

One task is to distinguish between the subjunctive and indicative (also characterized as factual statements (Amirqulova, 2026)) (James, 1986). The subjective mood can be found in blogs, discussion forums, social media platforms (Duy, 2019) across various topics such as economic and medical distress (Goldberg et al., 2009) and seeking advice for travel, food, and shopping. Additionally, online reviews for a product/service tend to convey a sentiment (positive/negative). Previous work stop at the sentiment level, whereas the subjunctive mood looks more fine-grained at the actual review for any potential production recommendations (Negi and Buitelaar, 2015; Duy, 2019).

Proposed Project: In this work, we will perform binary classification on the datasets from the Semeval 2019 Task 9 - SubTask A - Suggestion Mining from Online Reviews and Forums (Negi et al., 2019; Anderson and Hamilton, 2019), YouTube (Anderson and Hamilton, 2019), Reddit (Anderson and Hamilton, 2019), and miscellaneous (Duy, 2019). We will use machine learning (ML) models and open-source large language models (LLM) to classify. We will also use ML interpretability with shap (SHAP Developers, 2024) and LLM prompt type chain-of-thought for analysis on why the model performed how it does. We will also use the standard classification metrics with accuracy, precision, recall, f1, auc-roc, auc-pr.

Potential Project:

- Find more datasets or I curate and annotate a dataset for other subjunctive classes (e.g., wish, possibility, desire, advice) or indicative. We could then perform multi-class classification.
- Explore how the subjunctive mood play a role in the predictive discourse with TOLSA (my work, although not published yet). More specifically, use my TOLSA-BERT model to classify the datasets above as TOLSA x non-TOLSA, where we get the cross-labels of TOLSA x Suggestion.
- Fine-tune BERT (Devlin et al., 2019) on the datasets above and perform out-of-domain suggestion classification on datasets I already have.
- Look at the syntactic or dependency structure of the datasets above.
- Curate a more fine-grained dataset with labels past, present, future subjunctive moods and see the relationship.
- Cluster (without using the ground truth) for a visual representation.

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